Project: Functional Treat

Description:

Create a Data Analyzer and Transformer program in Python that utilizes various functions to manipulate and analyze data in one-dimensional (1D) and two-dimensional (2D) arrays (using lists). The project must demonstrate proficiency in:

- Utilizing built-in functions
- Creating user-defined functions (UDF)
- Using *args, **kwargs, and __doc__
- Implementing function recursion
- Applying lambda (anonymous) functions
- Using the global keyword
- Returning multiple values
- Sorting and transforming list-based data collections

Objective:

The project should allow users to analyze and transform data in various ways by selecting from a menu-driven interface. It will consist of a range of functions that cover each topic.

Requirements:

User Interface (UI):

- Create a menu-driven interface to let users choose from different data transformation and analysis options.
- Allow users to exit the program from the main menu.

Data Input:

- Use a 1D or 2D list (array) as input, with an option for the user to specify values manually or use sample data.

Tasks and Functionalities:

- Implement the following functionalities:
  A. Built-in Functions:

- Demonstrate usage of built-in functions like len(), sum(), min(), max(), etc., to display basic statistics about the array data.

B. User-defined Functions (UDF):

- Create specific functions for each task, such as calculating the average, finding duplicates, or displaying unique values.

C. *args, **kwargs, and __doc__:

- Use *args to accept multiple values and display them.
- Use **kwargs to print key-value pairs as a summary of dataset characteristics.
- Use __doc__ in each function to provide a description and print this information in the main menu.

D. Recursion:

- Implement a recursive function to calculate the factorial of a number or the Fibonacci sequence, showcasing recursion in a data analysis context.

E. Lambda Function:

- Use a lambda function to filter out values above or below a threshold, specified by the user.
- Apply lambda functions in conjunction with map() and filter() to manipulate list data.

F. Global Keyword:

- Use a global variable to store a dataset summary (e.g., total number of values and overall mean) that can be accessed across different functions.

G. Return Multiple Values:

- Create a function that returns multiple statistics about the dataset, such as minimum, maximum, and average values, which are then displayed individually in the main program.

H. 1D and 2D Array (List):

- Allow the user to input a 1D list or a 2D list (nested list).
- Display the 2D list in a formatted grid structure.

I. Sorting Collection Data Types:

- Use the sort() method on a 1D list to arrange values in ascending or descending order.
- Use the sorted() function to sort rows in a 2D list, showcasing the difference between in-place sorting and returning a new sorted list.

Example Operations:

- Sum and average of the dataset.
- Sorting values and displaying them.
- Filtering values based on user input using lambda functions.
- Printing a summary of dataset characteristics using **kwargs.
- Displaying recursive calculation results (factorial or Fibonacci).

Documentation:

- Use __doc__ within each function to describe its purpose.
- Provide clear instructions within the menu interface.

Example Console Interaction:

```
Welcome to the Data Analyzer and Transformer Program

Main Menu:
1. Input Data
2. Display Data Summary (Built-in Functions)
3. Calculate Factorial (Recursion)
4. Filter Data by Threshold (Lambda Function)
5. Sort Data
6. Display Dataset Statistics (Return Multiple Values)
7. Exit Program
Please enter your choice: 1
```

Step 1: Input Data

```
Please enter your choice: 1

Enter data for a 1D array (separated by spaces):
34 12 56 78 43 21 90

Data has been stored successfully!
```

Step 2: Display Data Summary (Built-in Functions)

```
Please enter your choice: 2

Data Summary:
- Total elements: 7
- Minimum value: 12
- Maximum value: 90
- Sum of all values: 334
```

```
- Average value: 47.71
```

Step 3: Calculate Factorial (Recursion)

```
Please enter your choice: 3

Enter a number to calculate its factorial: 5

Factorial of 5 is: 120
```

Step 4: Filter Data by Threshold (Lambda Function)

```
Please enter your choice: 4

Enter a threshold value to filter out data above this value:
50

Filtered Data (values >= 50):
56, 78, 90
```

Step 5: Sort Data

```
Please enter your choice: 5

Choose sorting option:
1. Ascending
2. Descending

Enter your choice: 1

Sorted Data in Ascending Order:
12, 21, 34, 43, 56, 78, 90
```

Step 6: Display Dataset Statistics (Return Multiple Values)

```
Please enter your choice: 6

Dataset Statistics:
- Minimum value: 12
- Maximum value: 90
- Sum of all values: 334
- Average value: 47.71
```

Step 7: Exit Program

```
Please enter your choice: 7

Thank you for using the Data Analyzer and Transformer Program. Goodbye!
```

Instructions:

Task Completion: Ensure that you attempt all the assigned tasks given to you as part of the exam. Complete each task to the best of your ability, following the instructions provided.

Assumptions: Make suitable assumptions wherever necessary, based on the requirements and instructions provided. Document any assumptions made in your project documentation or README.md file.

GitHub Repository: Create a GitHub repository to host your project. Upload your project files, including source code, and documentation to the repository. Ensure that you provide a clear and descriptive README.md file.

No Copying: Do not copy code or any other content from your classmates or any other source. Plagiarism is strictly prohibited and can result in severe consequences, including academic penalties. Ensure that all the code and content in your project are original and properly attributed to the appropriate sources, if applicable.

Submission: Once you have completed your project and uploaded it to your GitHub repository, submit the GitHub repository link to your instructor or as instructed. Double-check that your repository is properly organized and includes all the required files as per the instructions provided.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!